Resilience Academy

SDI Training for ICT Professionals

CRD Installation with Docker

Setting Up Your Own CRD Instance

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Day 2 | 30 Minutes (Lecture) | crd.resilienceacademy.ac.tz

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Session Overview: We Will Install CRD on Your Computer

What is about to happen:

In the practical session that follows this lecture, you will install a **complete copy of CRD** on your laptop using Docker.

Why install locally?

- You can **practise freely** — upload, delete, break things without worry
- You learn how the system **actually works** behind the scenes
- When your institution deploys CRD, you will **understand the process**
- No internet needed once installed

Step 1: Clone (copy the code)

↓

Step 2: Configure (set 3 settings)

↓

Step 3: Build (Docker creates everything)

↓

Step 4: Start (launch the platform)

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What Will Happen Step by Step

Step 1: Clone the Repository

- "Clone" means **download a copy** of all the CRD source files from GitHub
- One command, takes about 1–2 minutes
- Creates a folder called `crd` on your computer

Step 2: Configure the `.env` File

- The `.env` file is like a **settings form** — you fill in 3 values
- We provide a ready-made template; you just edit it
- Details on the next slide

Step 3: Build with Docker Compose

- Docker reads a recipe file (`docker-compose.yml`) and **builds all the pieces**: database, web server, map server, etc.
- Like assembling flat-pack furniture — Docker follows the instructions automatically
- Takes 10–15 minutes — be patient!

Step 4: Start the Services

- One command launches **all 7+ containers** together
- Takes 3–5 minutes for everything to be

The 3 Key Settings You Need to Know

The `.env` file controls how CRD behaves. Think of it as the **registration form** for your new CRD instance.

Setting 1: SITEURL

What it does: Tells CRD its own address

For local install:

`SITEURL=http://localhost/`

For a server:

`SITEURL=https://crd.yourorg.go.tz/`

Analogy: Like writing your home address on a form

Setting 2: ADMIN_PASSWORD

Setting 3: SECRET_KEY

What it does: A long random string that keeps your site secure

Example: `SECRET_KEY=a8k3j...x9m2p`

Like a unique lock combination — never share it

We provide a pre-generated one; just leave it as-is.

Important

The `.env` file is a plain text file. You edit it with any text editor (Notepad, nano, VS

What to Expect: Timing and Patience

Building takes time — this is normal!

Cloning: 1–2 min
Downloading source code from GitHub

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Configuring: 2–3 min
Editing the .env file

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Building: 10–15 min
Docker downloads & assembles everything

↓

Starting: 3–5 min
Services boot up one by one

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Verification: 1 min

During the build, you will see:

- Lots of text scrolling in the terminal — this is Docker downloading components and building images
- Progress bars and percentages
- Occasional warnings (yellow text) — usually safe to ignore
- The process finishes when you see a command prompt again

How to know it worked:

1. Open your web browser
2. Go to `http://localhost`

Common Problems and What to Do

Problem: "Port already in use"

What it means: Another application is using port 80 (the web port).

Solution:

- Close Skype, Apache, IIS, or any other web server
- On Windows: open Task Manager and stop the conflicting process
- Or change the port in `.env`:
  `HTTP_PORT=8080`
  then visit `http://localhost:8080`

Problem: "Docker is not running"

What it means: Docker Desktop has not been started.

Solution:

- Look for the Docker whale icon in your system tray
- If not there, open Docker Desktop from your applications
- Wait until it says "Docker is running" (green icon)
- Then try the command again

Summary: Ready for the Practical

What you now know:

- ✓ The installation has **5 clear steps**: clone, configure, build, start, verify
- ✓ You only need to set **3 values** in the `.env` file
- ✓ Building takes **10–15 minutes** — this is normal
- ✓ Success means seeing CRD at `http://localhost`
- ✓ Common problems have **simple solutions**

Quick checklist:

- ☐ Docker Desktop running
- ☐ Internet connection
- ☐ 10 GB free disk space
- ☐ Terminal/PowerShell open
- ☐ Practical guide ready

The Practical Guide

Your printed practical guide has **every command numbered**. You do not need to memorise anything